

Cost Function

$$J(\vec{w}, b) = -\frac{1}{m} \sum_{i=1}^m \left[y^{(i)} \log(f_{\vec{w}, b}(\vec{x}^{(i)})) + (1 - y^{(i)}) \log(1 - f_{\vec{w}, b}(\vec{x}^{(i)})) \right]$$

$$+ \left[\frac{\lambda}{2m} \sum_{j=1}^n w_j^2 \right]$$

Gradient Descent

repeat $\hookrightarrow$

$$w_j = w_j - \alpha \frac{\partial}{\partial w_j} J(\vec{w}, b)$$

$$b = b - \alpha \frac{\partial}{\partial b} J(\vec{w}, b)$$

$\hookrightarrow$ simultaneous update

return

where,

$$\frac{\partial}{\partial \omega_j} J(\vec{\omega}, b)$$

$$= \frac{1}{m} \sum_{i=1}^m [(f(\vec{\omega}, b)(\vec{x}^{(i)}) - y^{(i)})^2] \cdot x_j^{(i)} + \boxed{\frac{\lambda}{m} \omega_j}$$

Here, $f(\vec{\omega}, b)(\vec{x}^{(i)}) = \frac{1}{1 + e^{-(\vec{\omega} \cdot \vec{x} + b)}}$